

How free fatty acids inhibit glucose utilization in human skeletal muscle.



Rat muscle studies suggest competition between free fatty acids (FFA) and glucose for oxidation, resulting in glucose-6-phosphate accumulation. However, FFA decrease glucose-6-phosphate in human skeletal muscle, indicating direct inhibition of glucose transport/phosphorylation. This mechanism could redirect glucose from muscle to brain during fasting and explain the insulin resistance associated with high-lipid diets and obesity.